Exact Synthesis of Two-Qubit Clifford+CS Circuits with a Near-Minimal Number of K and CS Gates

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We give a method for the decomposition of an arbitrary two-qubit Clifford+CS operator (equivalently a 4×4 unitary with entries in $\mathbb{D}[i] = \mathbb{Z}[\frac{1}{2}, i]$), into a Clifford+CS circuit. Given unitary U, let k be the least natural number such that entries of $(1+i)^k U$ are all Gaussian integers, then this method decomposes U, upto a global phase, into a circuit of length at most $20k + 9$, using at most $2k$ K (i.e. a scaled H) gates and at most $2k + 1$ Controlled-S gates, without using any ancilla. Moreover, the number of K gate is at most 2 times the optimal, the number of *CS* gate is at most 4 times the optimal. In expriment, this method acheives the optimal CS-count always, and on average a 19% K-count reduction than the best-known. The method used is similar to the work of Russell [39] for the Clifford+T gate set but with a more concrete bound analysis (e.g. bound 20k+9 v.s. $O(k)$), where the ring in use is $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$ which is more complicated than and contains the ring here $\mathbb{Z}[\frac{1}{2}, i]$.

I. INTRODUCTION

In quantum computation, the decomposition of unitary operators into gates from a fixed universal set is an important problem [37]. Like the Clifford+T gate set, the Clifford+Controlled-S is also universal. We study the exact synthesis 2-qubit Clifford+Controlled-S (the Controlled-S gate is abbreviated as CS). Previous work on two-qubit synthesis of Clifford+CS operators or Clifford+T operators mainly focus on minimizing the use of CS gate or T gate since they are more expensive than Clifford operators in certain scenerio. But works on T gate optimization [1, 5, 9, 10, 23, 27] suggested that the H gate is an "obstruction" to their T-optimization method. Quoting from [1] summarizes the reason: "It has been shown that T-depth optimizations can be implemented efficiently for circuits consisting only of T and CNOT gates and that H gates impede the optimization significantly".

We propose a method that produces circuits with a near-minimal number of the K (i.e., a scaled H) gates and CS gates, hence alleviate such "obstruction". Synthesis problem that aims to minimize the use of two different gates simultaneously is also of mathematical interest. One of the reason is that previous work on single-qubit synthesis of Clifford+T operators acheieved the minimum use of Hadamard and T gates [16, 29, 33], an extension of which to multi-qubit case has been an open problem.

A. Background

Consider the following unitary operators:

$$i,\ K = \omega^7 H = \frac{1}{1+i}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix},\ S = \begin{bmatrix}1 & 0\\ 0 & i\end{bmatrix},\ \text{and}$$

$$CZ = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & -1\end{bmatrix}.$$

Here, i the complex unit; $\omega = e^{\frac{i\pi}{4}}$ is the 8th roof of unity; H is the Hadamard gate, and K is a scaled version of H; S is sometimes called the *phase gate*, and CZ is the controlled-Z gate. When closed under multiplication, identities, and tensor products, these operators generate the *Clifford group* (possibly up to scalars, depending which scalars are included in the Clifford group; we only include the scalars ± 1 and $\pm i$ in our defintion; hence, the above gates generate the Clifford group precisely).

Every operator U obtained in this way is of size $2^n \times 2^n$ for some natural number n, and as usual, we say that U is an operator on n qubits. It is well-known that the group of n-qubit Clifford operators is finite for any given n (see, e.g., [40]), and therefore not universal for quantum computing. We can obtain a universal gate set by adding the controlled phase gate

$$CS = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & i\end{bmatrix}.$$

The resulting operators are called the Clifford+CS operators. Note that $CZ = CS^2$. We write $\mathcal{CLI}$ for the group of 2-qubit Clifford operators; $\mathcal{CCS}$ for the 2-qubit Clifford+CS operators; and $\mathcal{CIR}$ for the set of 2-qubit

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Clifford+CS circuits (a circuit is a finite sequence of gates). Each circuit $D \in \mathcal{CIR}$ implements a unique operator in $\mathcal{CCS}$, denoted by $[D]$. In other words, $[_]$ is the matrix semantic function from $\mathcal{CIR}$ to $\mathcal{CCS}$.

We use the following notations: we write $K_0 = K \otimes I, K_1 = I \otimes K$, and similarly for S_0 and S_1 where I is the 2×2 identity operator. We identify the scalar i with the 4×4-matrix $i\,(I \otimes I)$. In the notation for controlled gates, we use the convention that the target is the last index. For example, CX_{01} is a controlled X-gate with control qubit 0 and target qubit 1, also shorted as CX. CX_{10} is shorted as XC. The controlled S- and Z-gates are symmetric, in the sense that $CS_{jk} = CS_{kj}$ and $CZ_{jk} = CZ_{kj}$, and therefore the order of indices does not matter for them, so we just write CS and CZ. We also use the Z-gate, the X-gate, the controlled K-gate $CK = I \oplus K$, and the swap gate, denoted by Ex. They are related to the "primitive" gates introduced so far by $Z = S^2$, $X = KZKi$, $CX = K_1 CZ K_1 \cdot i$, $Ex = CX\,XC\,CX$, and

$$CK = CZ \cdot Z_0 S_0 Z_1 S_1 K_1 CS K_1 S_1 \cdot i. \tag{1}$$

Similar to CX gate, CK_{01} is shorted as CK and CK_{10} is shorted as KC. Note that by Equation 1, $CK \in \mathcal{CCS}$.

B. Related work

Operator synthesis can be done exactly or approximately. Here we only list a few of the closely related ones on exact synthesis.

1. *Single-qubit and single-qudit*

For single-qubit synthesis of Clifford+T operators, works [16, 29, 33] achieved the minimum use of both Hadamard and T gate. The work [18, 38] achieved the minimum use of T gate for single-qutrit synthesis of Clifford+T operators. The work [25] considered the single-qudit synthesis of Clifford+T operators. Forest et al. [12] generalized exact synthesis from Clifford+T to single-qubit Clifford-cyclotomic gate sets. Kliuchnikov and Yard [31] developed a general algebraic framework for exact synthesis over number rings, unifying several gate sets under one theory.

2. *Two-qubit*

Two works on two-qubit synthesis have guaranteed T-optimality or CS-optimality: 2-qubit Clifford+T synthesis [32] and 2-qubit Clifford+CS synthesis [19]. The first gave an algorithm that is near T-optimal (the worst case uses at most 10 times the minimum possible T gates). The latter gave an algorithm that is CS-optimal. Both haven't considered the H-count optimization, but we can deduce, for the latter work, that their H-count enjoys the same property as ours: the K-count is at most 2 times the optimal. Glaudell et al. [17] later gave a high-performance exact synthesis framework for two-qubit Clifford+T circuits that optimizes T-count exactly, using meet-in-the-middle with algebraic canonicalization and lookup tables. The above 3 works depends on the exceptional isomorphism between two Lie groups $SU(4) \cong \mathrm{spin}(6)$. Our work is about the CS-optimality and H-optimality of 2-qubit Clifford+CS synthesis, but without using this exceptional isomorphism. Our work use the similar "residue" analysis method as in Russell [39], but over a simpler ring $\mathbb{Z}[\frac{1}{2}, i]$ contained in their ring $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$. Given a unitary U, Russell's method output circuit of gate count $O(k)$, where k is the least natural number such that entries of $(1+\omega)^k U$ are in the ring $\mathbb{Z}[\omega]$.

3. *Multi-qubit and multi-qutrit*

Multi-qubit synthesis of Clifford+T operators is proposed by Giles and Selinger [15]. The algorithm's worst case gate count is $O\left(3^{2^n} nk\right)$. Later work [28] improved the gate count to $O\left(4^n nk\right)$. Glaudell et al. later gave the exact multi-qubit synthesis of several subclasses of Clifford+T operators [4] such as the class of operators generated by $\{X, CX, CCX, H\}$, known as the Toffoli+Hadamard operators. Amy et al. adapted the method used in [28] to the Toffoli+Hadamard operators achieving a similar improvement on gate count [3]. All the works on multi-qubit synthesis introduced so far haven't considered T-count optimality. Kliuchnikov and Schönnenbeck [30] recently developed a optimal search algorithm for multi-qubit exact synthesis using A^* search. In particular, for two-qubit circuits, the runtime is proportional to the T-count. The work [2, 11] considered exact synthesis of multi-qubit Clifford+Cyclotomic operators. The work [21, 26] considered exact synthesis of multi-qutrit Clifford+Cyclotomic operators. The work [20] proved that qutrit metaplectic gates are a subset of Clifford+T. The works [13, 14, 34–36] proposed heuristic algorithms that acheive kinds of non-Clifford gate optimality.

II. SOME ALGEBRA

A. Rings

As usual, we write $\mathbb{N} = \{0, 1, \ldots\}$ for the set of natural numbers. We also write $\mathbb{Z}$, $\mathbb{R}$, and $\mathbb{C}$ to denote the rings of integers, real numbers, and complex numbers, respectively. We write $z^\dagger$ for the complex conjugate of a complex number. Recall that in any ring, a *unit* is an invertible element.

Let $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ be the ring of Gaussian

integers. It is a Euclidean domain[6], and therefore, division with remainder, greatest common divisors, divisibility, and the concept of a prime (i.e. a non-unit that cannot be written as the product of two non-units) all can be used in $\mathbb{Z}[i]$. Let $\gamma = 1 + i$. It is well-known that γ is a Gaussian prime [24]. Note that γ is a divisor of 2, and γ^2 and 2 divide each other. Using Euclidean division, it is further easy to check that

$$\mathbb{Z}[i]/(\gamma) = \{0, 1\}, \quad \mathbb{Z}[i]/(2) = \mathbb{Z}[i]/(\gamma^2) = \{0, 1, i, 1+i\},$$

$$\text{and} \quad \mathbb{Z}[i]/(\gamma^3) = \{\pm 1, \pm i, 0, 1 + i, 1 - i, 2\}.$$

Note that the above is a slight abuse of notation. Normally, $\mathbb{Z}[i]/(\gamma)$ denotes the quotient ring of $\mathbb{Z}[i]$ modulo the ideal (γ) whose elements are residues nomrally denoted by $x \bmod(\gamma)$ for $x \in \mathbb{Z}[i]$. Here we omit the mod (γ) for simlicity. Note that $\{0, 1\}$ etc. are complete sets of residue classes and also the reminders from the Euclidean division of x divided by γ.

Let $\mathbb{D} = \mathbb{Z}[\frac{1}{2}] = \left\{\frac{a}{2^k} \mid a \in \mathbb{Z}, k \in \mathbb{N}\right\}$ be the ring of *dyadic fractions*, i.e., fractions whose denominator is a power of 2. Consider the ring $\mathbb{D}[i] = \{r + si \mid r, s \in \mathbb{D}\} = \mathbb{Z}[\frac{1}{2}, i]$. For every $t \in \mathbb{D}[i]$, there exists some natural number k such that $\gamma^k t \in \mathbb{Z}[i]$. This motivates the following definition.

Definition II.1. Let $t \in \mathbb{D}[i]$. A natural number $k \in \mathbb{N}$ is called a *denominator exponent* for t if $\gamma^k t \in \mathbb{Z}[i]$. The least such k is called the *least denominator exponent* of t, denoted by $\mathrm{lde}(t)$. Equivalently, the least denominator exponent of t is the smallest $k \in \mathbb{N}$ such that t can be written in the form $\frac{s}{\gamma^k}$, where $s \in \mathbb{Z}[i]$.

More generally, we say that k is a denominator exponent for a vector or matrix if it is a denominator exponent for all of its entries. The least denominator exponent for a vector or matrix is therefore the least k that is a denominator exponent for all of its entries.

Remark II.2. By enumerating all 2-qubit Clifford operators, we have that the lde of a 2-qubit Clifford operator is at most 2.

B. Residue rings

It is well-known that

$$\mathbb{Z}[i]/(\gamma^n) = \left\{\sum_{i=0}^{n-1} b_i \gamma^i \bmod (\gamma^n) \,|\, b_i \in \{0, 1\}\right\}.$$

We give an elmentary proof by induction on n. Note that similar proof is used for proving the case for $\mathbb{Z}[\sqrt{2}]/(\sqrt{2}^n)$ by Clement[8]. When $n = 1$, the claim obviously holds. Suppose this is true for n, i.e., for any $x \in \mathbb{Z}[i]$, x is congruent to some $\sum_{i=0}^{n-1} b_i \gamma^i \bmod (\gamma^n)$, that is

$$x = \sum_{i=0}^{n-1} b_i \gamma^i + k\gamma^n \text{ for some } k.$$

Since $k = b_n + k'\gamma$ for some $b_n \in \{0, 1\}$, substituting k in and some rewrite gives

$$x = \sum_{i=0}^{n} b_i \gamma^i + k'\gamma^{n+1}, \text{ i.e., } x \equiv \sum_{i=0}^{n} b_i \gamma^i \bmod (\gamma^{n+1}).$$

For simplicity, we omit the mod (γ^n) in $\sum_{i=0}^{n-1} b_i \gamma^i \bmod (\gamma^n)$. Since $b_i \in \{0, 1\}$, we then write binary string $(b_0 b_1 ...)_2$ for the summation, and the parenthesis and subscript $_2$ can be omitted if we can infer that $b_0 b_1 ...$ is a residue in binary format from the context.

Moreover, residue classes are distinct, i.e. if $b_0 b_1 ... b_{n-1}$ and $b'_0 b'_1 ... b'_{n-1}$ are equivalent, i.e., $\sum_{i=1}^{n-1} b_i \gamma^i - \sum_{i=1}^{n-1} b'_i \gamma^i = k\gamma^n$ for some k, then $b_i = b'_i$. Modding (γ) on both side gives $b_0 = b'_0$. Then cancelling b_0 and b'_i, and dividing both side by γ give

$$\sum_{i=1}^{n-1} b_i \gamma^{i-1} - \sum_{i=1}^{n-1} b'_i \gamma^{i-1} = k\gamma^{n-1}$$

Then modding (γ) on both side agian gives $b_1 = b'_1$. Repeat this process proves $b_i = b'_i$.

We write $\rho_n(x)$ for the cannonical projection from $\mathbb{Z}[i]$ to n-digit residues. From the proof, we see $\rho_n(x)$ is "stable" w.r.t increase of n, i.e., the first n digits of $\rho_{n+1}(x)$ is $\rho_n(x)$. Note that ρ_2 is "stable" under taking complex conjugation, i.e., $\rho_2(x^\dagger) = \rho_2(x)$.

We sometimes write $x \equiv_{\gamma^n} y$ as an abbreviation for $x \equiv y \bmod \gamma^n$. Modulo γ^3, we have

x	$\rho_3(x)$	x	$\rho_3(x)$	x	$\rho_3(x)$	x	$\rho_3(x)$
1	100	-1	101	i	111	$-i$	110
0	000	$1+i$	010	$1-i$	011	2	001

The (mod γ^n) residue classes, as the quotient ring, is a commutative ring. In particular, modulo γ^2, the addition of the ring coincides with the bitwise exclusive or (denoted by $\oplus$) operation; $(10)_2$ is the ring identity; $(00)_2$ the zero of the ring; multiplication by $(01)_2$ is the right shift operation; by distributivity $11 \cdot b_0 b_1 = 10 \cdot b_0 b_1 + 01 \cdot b_0 b_1 = b_0(b_1 \oplus b_0)$. Note that since $\rho_2(i) = 11$, if $\rho_1(x) = 1$, then either $\rho_2(x) = 10$ or $\rho_2(ix) = 10$. Similarly, modulo γ^3, the addition is

$$b_0 b_1 b_2 + b'_0 b'_1 b'_2 = (b_0 \oplus b'_0)(b_1 \oplus b'_1)(b_2 \oplus b'_2 \oplus (b_0 \cdot b'_0)),$$

where the $\cdot$ is the multiplication of the ring $\mathbb{Z}[i]/(\gamma)$, equivalently the "and" of two bits. The ring identity is 100, zero is 000. Multiplication of 010 is a right shit by 1 digit and multiplication of 001 is a right shit by 2 digits. The rest of the multiplication table can be obtained using distributativity. The additive inverse is

$$-(b_0 b_1 b_2) = b_0 b_1 (b_2 \oplus b_0).$$

By looking at the ρ_3 table, we see ρ_3 is not "stable" under taking complex conjugation, but instead, if $\rho_3(x) = abc$, then $\rho_3(x^\dagger) = ab\bar{c}$. Since $\rho_3(i) = 111$, if $\rho_1(x) = 1$, then there exists $k \in \{0, 1, 2, 3\}$ such that $\rho_3(i^k x) = 100$.

Definition II.3. For $x \in \mathbb{Z}[i]$, if $\rho_1(x) = 0$, we say x is *even*, otherwise *odd*.

For example, $2+3i$ and $5+2i$ are odd, and $2+4i$ and $1+3i$ are even. In particular, in our context, by an even Gaussian integer we mean one that is divisible by γ, not necessarily by 2. However if $x \in \mathbb{Z}$, then x is even (odd) as an integer if and only if x is even (odd) as a Gaussian integer. Therefore, our definition of parity extends the usual one on the integers. The following lemmas are straightforward.

Lemma II.4. *Let $\alpha = a + bi \in \mathbb{Z}[i]$. Then α is odd if and only if $\|\alpha\|^2 = a^2 + b^2$ is an odd integer.*

Lemma II.5. *If $\alpha \in \mathbb{Z}[i]$ is odd, then $\alpha \equiv \pm 1, \pm i \pmod{\gamma^3}$. In other words, $\alpha \equiv i^e \pmod{\gamma^3}$ for some $e \in \{0,1,2,3\}$. Moreover, $\alpha \equiv i^e \pmod{\gamma^2}$ for some $e \in \{0,1\}$.*

Lemma II.6. *If $\alpha, \beta \in \mathbb{Z}[i]$ satisfy $\alpha\gamma \equiv_{\gamma^{n+1}} \beta\gamma$, then $\alpha \equiv_{\gamma^n} \beta$.*

Let k be a denominator exponent (not necessarily the least) for $x \in \mathbb{D}[i]$, we define the residue function $\rho_n^k(x) = \rho_n(\gamma^k x)$. We call $\rho_n^k(x)$ the (n,k) residue of x. If k is a denominator exponent for a vector or matrix A, $\rho_n^k(A)$ is defined entry-wise.

Lemma II.7. *For $t \in \mathbb{D}[i]$, if $k = lde_\gamma(t) > 0$, then $\gamma^k t$ is odd.*

Lemma II.8. *For any $x \in \mathbb{D}[i]$ with denominator exponent k, $\rho_2^k(x^\dagger) = \rho_2^k(x)$ and $\rho_3^k(x^\dagger) = \rho_2^k(x) + 001$.*

C. Action on residues

Suppose $\rho_2(x) = b_0 b_1$, then $\rho_2(ix) = b_0(b_1 \oplus b_0)$. We say i acts on $b_0 b_1$ by sending it to $b_0(b_1 \oplus b_0)$. Given an n residue $b_0 b_1 ... b_{n-1}$, a right shift of it is an $n+1$ residue $0 b_0 b_1 ... b_{n-1}$. Given an $n+1$ residue $0 b_0 b_1 ... b_{n-1}$, a left shift is $b_0 b_1 ... b_n$. Note that we choose to only allow "well-defined" shifts in the sense no information is lost and no unknown is introduced. Note that the usual right shift is used previously when explaining the multiplication of $\mathbb{Z}[i]/(\gamma^2)$ and $\mathbb{Z}[i]/(\gamma^3)$. We use $LS(x)$ for the left shift of x and RS for the right shift.

Lemma II.9. *For $t \in \mathbb{D}[i]$, k a denominator exponent of t, we have*

$$\rho_{n+1}^{k+1}(t) = \rho_{n+1}^k(\gamma t) = RS(\rho_n^k(t)) \text{ and}$$

and if $\gamma^k t$ is even,

$$\rho_n^{k-1}(t) = \rho_n^k(t/\gamma) = LS(\rho_{n+1}^k(t)).$$

Given $x, y \in \mathbb{D}[i]$ with lde $k > 0$, we know $\rho_2^k(x) = 1b$ and $\rho_2^k(y) = 1b'$ for some $b, b' \in \{0,1\}$. So $\rho_2^k(x+y) = 1b + 1b' = 0(b \oplus b')$. Then $\rho_1^{k-1}(x+y) = LS(\rho_2^k(x+y)) = b \oplus b'$. Similarly, $\rho_1^{k-1}(x-y) = LS(\rho_2^k(x-y)) = LS(1b - 1b') = LS((1-1)(b-b')) = b \oplus b'$ (subtraction in $\mathbb{Z}/(2)$ coincides with $\oplus$, in other words the additive inverse of b is itself). Note that if $b \oplus b' = 1$, then $\mathrm{lde}(x \pm y)$ decreases by 1, and if $b \oplus b' = 0$, $\mathrm{lde}(x \pm y)$ decreases by at least 2 if $k \geq 2$. Since dividing t by γ obviously increases $\mathrm{lde}(t)$ by 1, we have for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho_3^k(x) = 10b$ and $\rho_3^k(y) = 10b'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \rho_2^k(\begin{bmatrix} (x+y)/\gamma \\ (x-y)/\gamma \end{bmatrix}) = LS(\rho_3^k(\begin{bmatrix} x+y \\ x-y \end{bmatrix})) = \begin{bmatrix} 0(b \oplus b' \oplus 1) \\ 0(b \oplus b') \end{bmatrix}.$$

So we have $\mathrm{lde}(K \begin{bmatrix} x \\ y \end{bmatrix}) = k-1$ and one entry of it has lde $k-1$ and the other at most $k-2$ if $k > 2$.

In general for $x, y \in \mathbb{D}[i]$ with lde $k > 0$ and $\rho_3^k(x) = 1 b_0 b_1$ and $\rho_3^k(y) = 1 b_0' b_1'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \rho_2^k(\begin{bmatrix} (x+y)/\gamma \\ (x-y)/\gamma \end{bmatrix}) = LS(\rho_3^k(\begin{bmatrix} x+y \\ x-y \end{bmatrix})) = \begin{bmatrix} (b_0 \oplus b_0')(b_1 \oplus b_1' \oplus 1) \\ (b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

If $\rho_3^k(x) = 0 b_0 b_1$ and $\rho_3^k(y) = 0 b_0' b_1'$,

$$\rho_2^k(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} (b_0 \oplus b_0')(b_1 \oplus b_1') \\ (b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

If $\rho_3^k(x) = 1 b_0 b_1$ and $\rho_3^k(y) = 0 b_0' b_1'$,

$$\rho_3^{k+1}(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b_0')(b_1 \oplus b_1') \\ 1(b_0 \oplus b_0')(b_1 \oplus b_1') \end{bmatrix}.$$

If $\rho_3^k(x) = 0 b_0 b_1$ and $\rho_3^k(y) = 1 b_0' b_1'$,

$$\rho_3^{k+1}(K \begin{bmatrix} x \\ y \end{bmatrix}) = \begin{bmatrix} 1(b_0 \oplus b_0')(b_1 \oplus b_1') \\ 1(b_0 \oplus b_0')(b_1 \oplus b_1' \oplus 1) \end{bmatrix}.$$

III. SPECIAL UNITARIES

A. Permutation matrices

Let $a_0 a_1 a_2 a_3$ be a permutation of $\{0,1,2,3\}$. The permutation matrix $P_{a_0 a_1 a_2 a_3}$ is the unitary matrix that maps e_i to e_{a_i}, where $\{e_i\}$ is the standard basis. All 24 permutation matrix are Clifford operators and they can be implemented by circuits of lenght at most 3, see Table I.

P	circuit	P	circuit	P	circuit
0123	identity	1023	$X_1 \cdot CX$	2103	$X_0 \cdot XC$
1203	$Ex \cdot X_0 \cdot XC$	2013	$Ex \cdot X_1 \cdot CX$	0213	Ex
3210	$X_1 \cdot X_0$	2310	$X_0 \cdot CX$	2130	$Ex \cdot X_1 \cdot XC$
3120	$Ex \cdot X_1 \cdot X_0$	1320	$Ex \cdot X_0 \cdot CX$	1230	$X_1 \cdot XC$
3012	$XC \cdot X_1$	0312	$Ex \cdot XC$	0132	CX
3102	$Ex \cdot CX \cdot X_0$	1302	$Ex \cdot X_0$	1032	X_1
3021	$Ex \cdot XC \cdot X_1$	0321	XC	0231	$Ex \cdot CX$
3201	$CX \cdot X_0$	2301	X_0	2031	$Ex \cdot X_1$

TABLE I. Circuit implementation of 24 permutation matrices.

B. Diagonal unitary matrices

The only diagonal unitary matrices with entries in $\mathbb{D}[i]$ (and $\mathbb{Z}[i]$) are of the form

$$\mathrm{diag}(i^a, i^b, i^c, i^d), \text{ where } a, b, c, d \in \{0, 1, 2, 3\}.$$

They can all be implemented by Clifford+CS circuits with length of at most 6 upto a global phase i^k, using at most 1 CS gate. This because

$$Z_0^{b_0} Z_1^{b_1} S_0^{b_2} S_1^{b_3} CZ^{b_4} CS^{b_5} \cdot i^k, b_i \in \{0, 1\}, k \in \{0, 1, 2, 3\},$$

are all distinct and diagonal, and it is of cardinality 4^4 which equals to the cardinality of the set of diagonal unitary matices.

C. Generalized permutation unitary matrices

A generalized permutation unitary matrix with entries in $\mathbb{D}[i]$ is a matrix with the same nonzero pattern as a permutation matrix (there is exactly one nonzero entry in each row and each column), but the nonzero entry can be any number of unit length, i.e, powers of i. We will call such a matrix a generalized permutation. It is easy to see each 4×4 generalized permutation is product of a permutation matrix and a diagonal unitary matrix, hence can be implemented by a circuit of lenght at most 9 containing at most 1 CS gate, and no K gate. Also the group of generalied permutation is semi-direct product of the group of permutations and the group of diagonal unitaries. Then in particular, for any permuation matrix P and diagonal matrix D, there exists D' such that

$$PD = D'P$$

and the CS-count (see Definition IV.2) of D and D' are the same.

IV. THE EXACT SYNTHESIS ALGORITHM

The method used is similar to the work of Russell [39] for the ring $\mathbb{Z}[\frac{1}{\sqrt{2}}, i]$ which is more complicated than and contains the ring here $\mathbb{Z}[\frac{1}{2}, i]$. Such residue analysis method is widely used in exact synthesis, e.g., in [3, 15, 16, 19, 32]. Note that, comparing to the ad hoc use of residues in each work, we have given in Section II a uniform way of dealing with residues of different rings using binary residues, which, we conjecture, works for any ξ-ring proposed in [30].

We start with the main theorem.

A. Main theorem

Theorem IV.1. *For any $A \in U(\mathbb{D}[i])$ with lde k, there exists a circuit C implementing A, using at most $2k$ K gates, at most $2k+1$ CS gates, and at most $9(2k+1)+2k$ gates upto a global phase.*

The proof, given later, is constructive, so it gives an algorithm for computing such circuit.

B. Optimality

Definition IV.2. Let D be a Clifford+CS circuit. Recall that $[D]$ is the unitary operator it implements. Define

- $\mathrm{rkc}(D)$ = the number of K gate that appears in D.
- $\mathrm{kc}(D) = \min\{\mathrm{rkc}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- $\mathrm{rcs}(D)$ = the number of CS gate that appears in D.
- $\mathrm{cs}(D) = \min\{\mathrm{rcs}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- $\mathrm{rlen}(D)$ = the number of gate that D contains.
- $\mathrm{len}(D) = \min\{\mathrm{rlen}(C) \mid C \in \mathcal{CIR} \text{ and } [C] = [D]\}$.
- by abuse of notation, $\mathrm{cs}, \mathrm{kc}, \mathrm{len}$ are also defined on operators that are implementable by circuits.

We say D is K-optimal, CS-optimal, and shortest if $\mathrm{rkc}(D) = \mathrm{kc}(D)$, $\mathrm{rcs}(D) = \mathrm{cs}(D)$, and $\mathrm{rlen}(D) = \mathrm{len}(D)$ repectively. We call $\mathrm{rkc}(D)$ and $\mathrm{rcs}(D)$ the (raw) K-count and (raw) CS-count of D repectively, and $\mathrm{kc}(D)$, $\mathrm{cs}(D)$ the optimal K-count and the optimal CS-count of D respectively.

Remark IV.3. Note that this calling differs with the other convention in e.g. [22], where they would call, e.g., $\mathrm{cs}(D)$ the CS-count of D. In other words, the other convention has no definition for "raw" K-count (rkc) or CS-count (rcs), and the "count" they have is only optimal.

Any Clifford+CS circuit D is an alternating sequence of Clifford parts C_i and a CS gate

$$D = C_0 \cdot CS \cdot C_1 \cdot CS \cdot \ldots \cdot C_n. \tag{2}$$

By enumerating all 2-qubit Clifford circuit, we know C_i is implementable by at most 2 K gate and $\mathrm{lde}(C_i) \leq 2$. By subadditivity of lde, we have $\mathrm{lde}(D) \leq \mathrm{lde}(C_0) +$

$\mathrm{lde}(CS)+\mathrm{lde}(C_1)\ldots\mathrm{lde}(C_n) \leq 2(n+1)$, that is, $\mathrm{lde}(D) \leq (2(\mathrm{rcs}(D)+1)$ which implies

$$\mathrm{lde}(D) \leq 2(\mathrm{cs}(D)+1).$$

Suppose 2 is CS-optimal, i.e., $\mathrm{cs}(D) = n$, we have $\mathrm{rkc}(D) \leq 2(\mathrm{cs}(D)+1)$ which implies

$$\mathrm{kc}(D) \leq 2(\mathrm{cs}(D)+1).$$

Any Clifford+CS circuit D is also an alternating sequence of generalized permutation parts P_i and a K_1 gate (by replacing all appearences of K_0 by $Ex \cdot K_1 \cdot Ex$):

$$D = P_0 \cdot K_1 \cdot P_1 \cdot K_1 \cdot \ldots \cdot P_n. \tag{3}$$

Each generalized permutation is implementable by at most 1 CS gate and uses no K gate. So we have $\mathrm{cs}(D) \leq \mathrm{rkc}(D)+1$, which implies

$$\mathrm{cs}(D) \leq \mathrm{kc}(D)+1.$$

We also have, by subadditivity of lde, that $\mathrm{lde}(D) \leq \mathrm{rkc}(D)$, which implies

$$\mathrm{lde}(D) \leq \mathrm{kc}(D).$$

Theorem IV.1 outputs D' such that $\mathrm{rkc}(D') \leq 2\mathrm{lde}(D)$ and $\mathrm{rcs}(D') \leq 2\mathrm{lde}(D)+1$, which, together with the provious inequalities, implies

$$\mathrm{lde}(D) \leq \mathrm{kc}(D) \leq 2\mathrm{lde}(D) \tag{4}$$

$$\mathrm{kc}(D)/2-1 \leq \mathrm{cs}(D) \leq 2\mathrm{lde}(D)+1 \tag{5}$$

From above inequalities, our synthesis algorithm is at worst 2 times the optimal K-count, and at worst 4 times the optimal CS-count (experiments show it always acheives the optimal CS-count, and better K-count than best-known, see Section V). Note that Inequality 5 is similar to Propsition 6 of [19], but slightly tigther.

1. *K optimality of Glaudell et al's algorithm*

The work of Glaudell et al. [19] didn't consider the K-count at all, but we can derive from their CS optimality that for a unitary A,

$$\mathrm{cs}(A) \leq \mathrm{kc}(A) \leq 2\,\mathrm{cs}(A)+2.$$

This is because the output of their algorithm is of the form Equation 2 with the CS-count being optimal and the Clifford in between CS gate has K-count 1 or 2. Take a C_i that is in between CS gate and call it E. First, $\mathrm{kc}(E) \neq 0$, becuase then E is also a generalized permutation which contains no CS gate because it is Clifford. So the two CS in $CSECS$ will cancell decreasing the CS-count, whcih contradicts the optmality. The cancelling can be shown by direct calculation since the generalized permutations are finite, or using the semi-direct product structure that the permuations and diagonals together form. We can show $\mathrm{kc}(E) \leq 2$ by enumerating all two-qubit Cliffords.

So Glaudell et al's algorithm enjoys the same property as ours: the K-count is at most 2 times worse than optimal.

Remark IV.4. Note that exmperiments show that most of the case $\mathrm{lde}(A) < \mathrm{cs}(A)$ for random generated A. So our bound $\mathrm{kc}(A) \leq 2\mathrm{lde}(A)$ is strictly better in practice.

C. Complexity

Since we only need finite many (not depending on the lde; proved later) operations to decrease the lde by 1, and the unitaries with lde 0 are generalized permutations, the time complexity is $O(\mathrm{lde}(U))$ for any input $U \in \mathcal{CCS}$.

D. Lemmas and proofs

We start with a number of lemmas. Since $\mathbb{Z}[i]$ and $\mathbb{D}[i]$ are subrings of $\mathbb{C}$, the usual properties of complex numbers, complex vectors, and complex matrices, such as complex conjugation (denoted by $_^\dagger$), inner product, norm, transpostion (denoted by $_^T$), and conjugate transposition (also denoted by $_^\dagger$) are naturally inherited. For example, for a vector $v \in \mathbb{D}[i]^n$, the *norm* of v is $\|v\| = \sqrt{v^\dagger v}$. Note that $\|v\|^2 = v^\dagger v \in \mathbb{D}$, since $v^\dagger v \in \mathbb{D}[i] \cap \mathbb{R}$. Similarly if $v \in \mathbb{Z}[i]$, then $v^\dagger v \in \mathbb{Z}$. A *unit vector* is a vector of norm 1.

If v is a vector, the notation v_j refers to its jth entry; entries are numbered starting from $j = 0$.

Lemma IV.5. *For any $A \in U(\mathbb{D}[i])$ with lde k, there exists permutation matrix L and R such that, $\rho_1^k(LAR)$ or $\rho_1^k(LAR)^T$ equals one of the following*

$$(i)\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{bmatrix},\quad (ii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (iii)\begin{bmatrix}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{bmatrix},$$

$$(iv)\begin{bmatrix}1&1&1&1\\1&1&1&1\\0&0&0&0\\0&0&0&0\end{bmatrix},\quad (v)\begin{bmatrix}1&1&0&0\\1&1&0&0\\1&1&1&1\\1&1&1&1\end{bmatrix},\quad (vi)\begin{bmatrix}1&1&1&1\\1&1&1&1\\1&1&1&1\\1&1&1&1\end{bmatrix}.$$

We also say A has such pattern and write $\mathrm{pat}(A)$ *for the pattern of A. Moreover, A has pattern (i) if and only if $\mathrm{lde}(A) = 0$, i.e., when A is a generalized permutation. A has pattern (iv), (v), and (vi) only if $\mathrm{lde}(A) > 1$.*

Before proving this lemma, we prove serveral useful lemmas.

Lemma IV.6. *Let v be a unit vector in $\mathbb{D}[i]^4$ with lde $k > 0$. There exists a generalized unitary permutation matrix P such that $\rho_1^k(Pv) = (1,1,0,0)^T$ or $(1,1,1,1)^T$, $\rho_2^k(Pv) = (10,10,01,00)^T$ or $(10,10,10,10)^T$ provided $k > 1$, and $\rho_2^1(Pv) = (10,10,00,00)^T$ provided if $k = 1$.*

Proof. Let $w = \gamma^k v \in \mathbb{Z}[i]^4$. We first show w has an even number of odd entries. We have

$$\sum_{j=0}^{3} \|w_j\|^2 = \|w\|^2 = |\gamma|^{2k}\|v\|^2 = (\gamma^\dagger\gamma)^k \equiv 0 \,(\mathrm{mod}\ \gamma).$$

Therefore, there are an even number of $j \in \{0, ..., 3\}$ such that $\|w_j\|^2$ is odd. It follows by Lemma II.4 that an even number of w_j are odd.

Then we can find a permutation matrix P such that $\rho_2^k(Pv) = (1a, 1b, 0c, 0d)^T$ or $(1a, 1b, 1c, 1d)^T$. For the later, by multiplying a diagonal matrix, i.e., multiplying i to the required entries, we can make the second digit all zero, i.e., $(10, 10, 10, 10)^T$.

For the former, by muliplying a diagonal matrix, we can make second digit of the odd entries zero, i.e., $(10, 10, 0c, 0d)^T$. Now applying CK (equivalently applying K to the pair of odd entries) to it, we get $(00, 01, 0c, 0d)^T$ or $(01, 00, 0c, 0d)^T$. Note that multiplication of a unitary and a unit vector is still a uinit vector, so is applying CK. So $(01, 00, 0c, 0d)^T$ or $(00, 01, 0c, 0d)^T$ is a residue of a unit vector with lde $k-1$, then its one-digit $k-1$ residue must of the form $(1,1,0,0)^T$ upto a permutation if $k > 1$, that is $c = 1$ and $d = 0$ upto a permutation.

If $k = 1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ for $l, m \in \{0, 1, 2, 3\}$ upto a permutation. □

Lemma IV.7. *Let u, v be two unit vectors in $\mathbb{D}[i]^4$ with lde $k > 0$ such that $u \perp v$, i.e., $u^\dagger v = 0$. Then there exists a generalized unitary permutation matrix Q such that $\rho_1^k(Q[u|v])$ equals*

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}, \text{ or } \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Proof. By Lemma IV.6, there can be only even number of 1s in each column. The only missing case is

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Then $\rho_1^k(u^\dagger v) = \sum_{j=0}^{3} \rho_1^k(\bar{u}_i v_i) = 1$ contradicts that $\rho_1^k(u^\dagger v) = \rho_1^k(0) = 0$. In other words, the number of indices such that at which the $(1, k)$ residue of u, v is 1 must be even. □

Proof of Lemma IV.5. Since the rows and columns of a unitary matrix are unit vectors and all row (column) vectors are perpendicular to each other. Lemma IV.6 and IV.7 forces the pattern to be (ii) - (vi) if $k > 0$. If $k = 0$, the only unitaries with lde 0 are the generalized permutations which has pattern (i). If $k = 1$, the only unit vectors are $\frac{1}{\gamma}(i^l, i^m, 0, 0)^T$ upto a permutation. So residue of such vectors cannot have four 1s. So Case (iv), (v), and (vi) only happens if the lde > 1. □

Lemma IV.8. *For any $A \in \mathcal{CCS}$ with lde $k > 0$, there exist generalized permutations L, R, L', R' such that one of the following holds:*

- *$lde(K_1LAR) \leq k - 1$;*
- *or $lde(LARK_1) \leq k - 1$;*
- *or $lde(L'K_1LARR'K_1) \leq k - 1$;*
- *or $lde(K_1L'K_1LARR') \leq k - 1$;*
- *or $lde(L'CKLAR) \leq k - 1$ (only happens when $lde(A) = 1$).*

Note that each $\leq$ above is actually $=$, but we don't need this. The proof proceeds by a case distinction over $\rho_1^k(A)$. Note that requiring $k > 0$ eliminates Case i. So start with Case ii.

1. Case ii

If $k > 1$, by Lemma IV.6, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 0b & 0\bar{b} \\ 01 & 0c & 0d & 0e \\ 00 & 0\bar{c} & 0f & 0g \end{bmatrix},$$

where $\bar{b} = b \oplus 1$ and b, c, d, e, f, g are indeterminates (later we use indeterminates in a residue matrix without declaring them). We claim $a = 0, b = c = 1$. Taking inner product of the first two cloumn gives

$$10 + 1a + 00 + 00 = 00 \Rightarrow a = 0.$$

Taking inner product of the second and third row gives $c = 1$; similarly, we have $b = 1$. Now pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 01 & 00 \\ 01 & 01 & 0d & 0e \\ 00 & 00 & 0f & 0g \end{bmatrix}.$$

Moreover, since $k > 1$ and the third and fourth row (column) is of lde $k - 1 > 0$, so $d = f$, $d = e$, $e = g$ by

Lemma IV.6. So $d = e = f = g$. Now pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 01 & 00 \\ 01 & 01 & 0d & 0d \\ 00 & 00 & 0d & 0d \end{bmatrix}.$$

Applying K_1S_1 on the left gives Case iv.

If $k = 1$, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 00 & 00 \\ 10 & 1a & 00 & 00 \\ 00 & 00 & 00 & 00 \\ 00 & 00 & 00 & 00 \end{bmatrix},$$

As before, we have $a = 0$. Apply $X_0 \cdot CK \cdot X_0$ on the left decreases the lde.

Remark IV.9. Note that the two generalized permutations we use to refine the pattern can be choosen to be CS-free. This is because the (2,k)-residue of the two rows (columns) of even entries are stable under multipicaton of i and we can introduce "dummy" i at a proper place to make a generalized permutation CS-free.

2. *Case iii*

If $k > 1$, by Lemma IV.6, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 0b & 0\bar{b} \\ 01 & 0c & 10 & 10 \\ 00 & 0\bar{c} & 10 & 1d \end{bmatrix},$$

Taking inner product of the first two cloumn gives $a = 0$. Taking inner product of the second and third row gives $c = 0$; similary, we have $b = 0$. Now pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 01 & 00 & 10 & 10 \\ 00 & 01 & 10 & 10 \end{bmatrix}.$$

Applying K_1S_1 on the left bring us to Case vi.

If $k = 1$. By Lemma IV.6, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 00 & 00 \\ 10 & 1a & 00 & 00 \\ 00 & 00 & 10 & 10 \\ 00 & 00 & 10 & 1d \end{bmatrix}.$$

As before, we have $a = 0, d = 0$. Applying K_1 on the left decrease the lde.

3. *Case iv*

This case only happens for $k > 1$. By Lemma IV.6, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1b & 1c \\ 01 & 0d & 0e & 0f \\ 00 & 0\bar{d} & 0\bar{e} & 0\bar{f} \end{bmatrix}.$$

Taking inner product of the first two cloumns $a = 0$; similarly we have $b = c = 0$. Since the third row is of lde $k - 1 > 0$, either $d = e = f = 1$ or one of d, e, f is 1 and the rest are 0. For the later, upto a permutation of columns, we may assume $d = 1, e = f = 0$. Then, for both cases, right multiplication of K_1 decrease the lde.

Remark IV.10. As in Remark IV.9, the left generalized permutation can be choosen to be CS-free.

4. *Case v*

This case only happens for $k > 1$. By Lemma IV.6, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 1a & 00 & 01 \\ 10 & 1b & 10 & 1f \\ 10 & 1c & 1e & 10 \end{bmatrix},$$

We claim $a = 0, b = c = e = 1$. Taking inner product of the first two rows gives $a = 0$. Taking inner product of the first and third row gives $b = 1$; similarly $c = 1$. Taking inner product of the second and third column gives $e = 1$; similarly $f = 1$. Now the pattern is

$$\begin{bmatrix} 10 & 10 & 01 & 00 \\ 10 & 10 & 00 & 01 \\ 10 & 11 & 10 & 11 \\ 10 & 11 & 11 & 10 \end{bmatrix},$$

Applying K_1 on the left bring us to Case iv.

5. *Case vi*

This case only happens for $k > 1$. By Lemma IV.6, upto multilying generalized permutations on the left and right, pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 1a & 1d & 1g \\ 10 & 1b & 1e & 1h \\ 10 & 1c & 1f & 1j \end{bmatrix},$$

Taking inner product of first two columns gives that a, b, c cannot be all 1. Upto a permutation, we may assume $a = 0$. Then $b = c$; similarly $d = g$.

If $b = c = 1$, and $d = g = 1$, pat(A) is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1h \\ 10 & 11 & 1f & 1j \end{bmatrix},$$

Taking inner product of the first and third column gives $f = \bar{e}$; similarly, $h = \bar{e}$ and $j = \bar{f}$. Now pat(A) is

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 11 & 11 \\ 10 & 11 & 1e & 1\bar{e} \\ 10 & 11 & 1\bar{e} & 1e \end{bmatrix}.$$

We prove this pattern cannot happen by looking at its (3,k)-residue. Upto multiplication of powers of -1 on the proper rows and columns, the (3,k)-residue refines to

$$\begin{bmatrix} 101 & 101 & 101 & 101 \\ 101 & 10x & 11u & 11p \\ 101 & 11y & 1ev & 1\bar{e}q \\ 101 & 11z & 1\bar{e}w & 1er \end{bmatrix}.$$

Taking inner product of the first column with the second and third column, we have

$$101 + 10x + 11y + 11z = 000 \implies x \oplus y \oplus z = 1,$$
$$101 + 11u + 1ev + 1\bar{e}w = 000 \implies u \oplus v \oplus w = 1.$$

Taking inner product of the second and third column gives

$$101 + 11(u \oplus \bar{x}) + (1ev + 01e + 00\bar{y}) + (1\bar{e}w + 01\bar{e} + 00\bar{z})$$
$$= 000 \implies x \oplus y \oplus z \oplus u \oplus v \oplus w = 1, \text{ a contradiction.}$$

If $b = c = 0$, taking inner product of the first and third row gives $e = h$, similarly, $f = j$. So pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 1d & 1d \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix},$$

Taking inner product of the first and third column gives $d \oplus e \oplus f = 0$, then at leat one of d, e, f is 0. Upto a permutation of rows, we may assume $d = 0$. Similar analysis applies if $d = g = 0$. So, for either case, upto permutations of rows and columns pat(A) refines to

$$\begin{bmatrix} 10 & 10 & 10 & 10 \\ 10 & 10 & 10 & 10 \\ 10 & 10 & 1e & 1e \\ 10 & 10 & 1f & 1f \end{bmatrix},$$

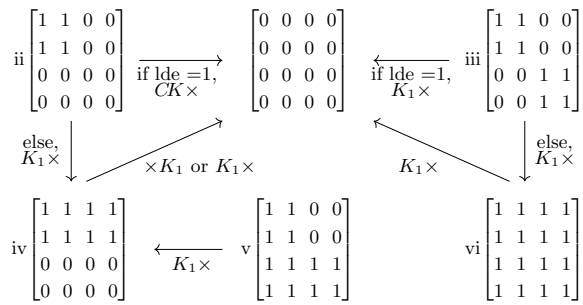

FIG. 1. Transitions between cases. The $K_1\times$ ($\times K_1$) annotated arrow means if A is of the pattern at the tail, then K_1LAR ($LARK_1$) is of the pattern at the head for appropritate choosen generalized permutations L and R. Similarly, for other annotated arrows. Note that CK decomposes to a circuit with two K gates as in Equation 1.

Taking inner product of the first and third column implies $e = f$. So left multiplication of K_1 reduces the lde.

Thus we have exhausted all cases. The transitions is summarized in Figure 1.

Proof of Theorem IV.1. Let $k = \text{lde}(A)$, by Lemma IV.8, by possibly merging adjacent generalized permutaions, there exist generalized permutations $L_1, L_2, ..., L_p, R_1, R_2, ..., R_q$ such that

$$\text{lde}(L_pK_1...L_2K_1L_1AR_1K_1R_2K_1..., R_q) = 0.$$

That is there exists a generalized permutation P s.t.

$$L_pK_1...L_2K_1L_1AR_1K_1R_2K_1..., R_q = P$$

Note the rkc of left-handside $\leq 2k$. Then we have

$$A = L_1^{-1}iK_1L_2^{-1}iK_1...L_p^{-1}PR_q^{-1}iK_1...R_2^{-1}iK_1R_1^{-1}.$$

Note that the rkc of the right handside is at most $2k$, after merging adjacent generalized permutaions, the number of generalized permutations is at most $2k + 1$. Each generalized permutation is implementable by at most 9 gates and at most 1 CS gate, upto a global phase. So we have a circuit that implements A, using at most 2k K gates, at most 2k +1 CS gates, and at most 9(2k +1)+2k gates upto a global phase. □

V. EXPERIMENT

We implemented our algorithm in Haskell and did a comparison with Glaudell et al.'s CS-optimal synthesis algorithm [19]. A unitary of given lde range is randomly generated and fed to both algorithms, and the output circuits are compared on metrics rcs, rkc. The output of their algorithm is of the form Equation 2 with the CS-count being optimal. We added one step to their algorithm to keep the K-count as small as possible: for each C_i, we replace it with an equivalent circuit of minimum K-count (could be 0,1,2). We generated randomly 1,346

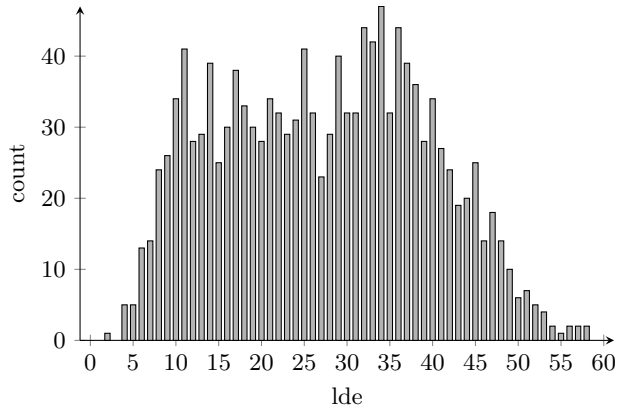

FIG. 2. Distribution of the least denominator exponent (lde). The lde ranges from 2 to 58 with mean 27 and median 28.

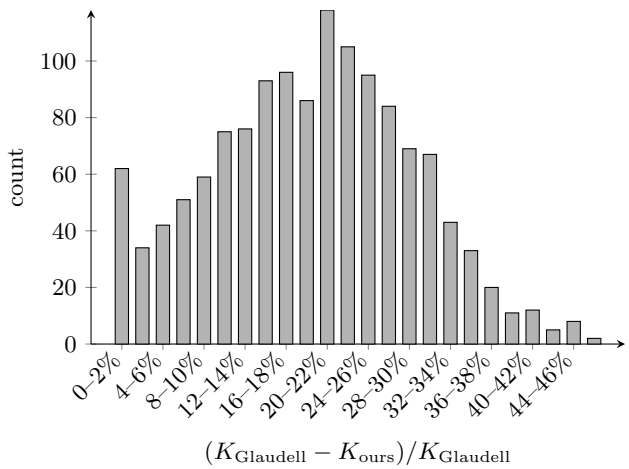

FIG. 3. Distribution of the relative K-count reduction $(K_{\text{Glaudell}} - K_{\text{ours}})/K_{\text{Glaudell}}$, binned in 2% increments. Our method improves on Glaudell et al.'s K-count in 95.5% of cases, with a mean reduction of 19.4%, a median of 19.8%, and a maximum of 47.7%.

two-qubit Clifford+CS operators with lde ranging from 2 to 58, available in VI. Figure 2 is the distribution of lde. Figure 3 is the distribution of the relative K-count reduction. From the data, we observe the following facts.

A. CS-count agreement

Across all 1,346 operators, our CS-count and Glaudell et al.'s CS-count are identical. So our algorithm is also CS-optimal in practice.

B. K-count improvement

Our algorithm produces a K-count less than or equal to that of Glaudell et al. in every case. It strictly improves the K-count in 95.5% of cases; see Figure 3.

C. Tighter bound of the optimal CS-count

The inequalities $\text{lde} - 2 \leq CS\text{-count} \leq K_{\text{ours}}$ hold for every operator in the dataset. In particular,

- $\text{lde} \leq CS$-count: 1,324 cases (98.4%),
- $\text{lde} > CS$-count: 22 cases (1.6%).

In the 22 cases, the deficit is usually 1, with the largest gap being 2 for $\text{lde} = 20$, $CS\text{-count} = 18$.

D. Source of improvement

Two tight linear relationships emerge from the data. First, the K-count of Glaudell et al. is well approximated by twice the CS-count:

$$K_{\text{Glaudell}} \approx 2 \cdot CS\text{-count}.$$

The residual $K_{\text{Glaudell}} - 2 \cdot CS\text{-count}$ lies in $\{-4, -3, -2, -1, 0, 1, 2\}$, with 93.5% of values in $\{-4, \ldots, 0\}$. Second, our K-count is well approximated by twice the lde:

$$K_{\text{ours}} \approx 2 \cdot \text{lde}.$$

The residual $K_{\text{ours}} - 2 \cdot \text{lde}$ lies in $\{-2, -1, 0\}$, with $\{-1, 0\}$ accounting for 80.5% of cases. These two relationships are also reflected in the Pearson correlations: the correlation between lde and K_{ours} is 0.9996, while the correlation between CS-count and K_{Glaudell} is 0.9991. Since $K_{\text{Glaudell}} \approx 2 \cdot CS\text{-count}$ and $K_{\text{ours}} \approx 2 \cdot \text{lde}$, the absolute K-count saving is approximately $2 \cdot (CS\text{-count} - \text{lde})$. In other words, the improvement is governed by the gap between the CS-count and the lde. The relative reduction percentage is roughly stable across lde values, hovering around 19–21% for lde between 5 and 40, and tapering off for very small or very large lde.

VI. CODE AND DATA AVAILABILITY

Code and data are available from [7].

VII. CONCLUSION AND FUTURE WORK

In conclusion, we gave an efficient and near-optimal algorithm for the exact synthesis of two-qubit Clifford+CS circuits from a given unitary matrix. In theory, the CS-count is at worest 4 times the optimal, but in practice, our method always acheives the optimal CS-count. In theory, the K-count is at worest 2 times the optimal, and in practice, our method always acheives the better K-count than the best-known.

The algorithm of Li et al. [32] outputs a circuit whose T-count is at worest 10 times the optimal T-count. We believe our method can be used to improve the bound 10.

VIII. ACKNOWLEDGEMENTS

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- creating four references [12, 17, 30, 31] and their one-line summary in the related work section.
- improving the wording of the paper.
- generating Haskell code: the ParseData.hs (for parsing data and drawing distribution diagrams) and CliffordCS.hs (the implementation of Glaudell et al.'s algorithm [19]).
- drafting Section VI, with proper guide from the authors (later revisions are made).

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